

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.10 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 Let T_r be the r^{th} term of a sequence, for $r = 1, 2, 3, \dots$. If $3T_{r+1} = T_r$ and $T_7 = \frac{1}{243}$, then the value of $\sum_{r=1}^{\infty} (T_r \cdot T_{r+1})$ is
(A) $\frac{9}{2}$ (B) $\frac{27}{8}$ (C) $\frac{81}{8}$ (D) $\frac{81}{4}$
- Q.2 If $S_n = \sum_{r=1}^n \sum_{t=0}^{r-1} \left(\frac{1}{6^n} {}^nC_r {}^rC_t 4^t \right)$, then value of l where $l = \sum_{n=1}^{\infty} (1 - S_n)$ is
(A) 5 (B) 3 (C) 4 (D) 6
- Q.3 If the number of fourteen digit numbers which can be formed by using the digits 1, 2, 3, 4, 7, 7, 7, ..., 7 (digit 7 is used 10 times) such that there are atleast two identical digits between distinct digits (distinct digits are 1, 2, 3 & 4) is $k \cdot 5!$ then k is
(A) 3 (B) 7 (C) 14 (D) none
- Q.4 How many factors of $25 \cdot 36 \cdot 52$ are perfect squares?
(A) 8 (B) 12 (C) 24 (D) 30
- Q.5 We have 19 identical gems available with us which are needed to be distributed among A, B and C such that A always gets an even number of gems. The number of ways this can be done is
(A) 72 (B) 92 (C) 100 (D) 110
- Q.6 An eight-digit telephone number consists of exactly two zeroes. One of the digits is repeated thrice. Remaining three digits are all distinct. If the first three digits (from left to right) are 987, then the probability of having only one 9, one 8 and one 7 in the telephone number, is
(A) $1/18$ (B) $1/20$ (C) $1/10$ (D) $5/47$
- Q.7 A bag contains 10 balls numbered from 0 to 9. The balls are such that the person picking a ball out of the bag is equally likely to pick anyone of them. A person picked a ball and replaced it in the bag after noting its number. He repeated this process 2 more times. The probability that the ball picked first is numbered higher than the ball picked second and the ball picked second is numbered higher than the ball picked third, is
(A) $72/100$ (B) $3/25$ (C) $4/5$ (D) $1/6$
- Q.8 Three consecutive letters are selected either from the word 'FANAND' or 'FAFANAND' so that these form a palindrome. The probability that they are taken from the word 'FANAND' is
(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{5}$ (D) $\frac{3}{7}$
- Q.9 A, B, C try to hit a target simultaneously but independently. Their respective probabilities of hitting the targets are $\frac{3}{4}, \frac{1}{2}, \frac{5}{8}$. The probability that the target is hit by A or B but not by C, is
(A) $\frac{9}{64}$ (B) $\frac{7}{8}$ (C) $\frac{21}{64}$ (D) $\frac{7}{32}$

- Q.10 Given two independent events, if the probability that exactly one of them occurs is $\frac{26}{49}$ and the probability that none of them occurs is $\frac{15}{49}$, then the probability of more probable of the two events is
- (A) $\frac{5}{7}$ (B) $\frac{2}{7}$ (C) $\frac{4}{7}$ (D) $\frac{6}{7}$

[PARAGRAPH TYPE]

Q.11 to Q.14 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 11 to 14

The seven digit number $N = a_1a_2a_3a_4a_5a_6a_7$ is divisible by 11 and sum of its digits equals 61.

- Q.11 Smallest digit that can be used will be
(A) 5 (B) 6 (C) 7 (D) 8
- Q.12 $a_1 + a_3 + a_5 + a_7$ is equal to
(A) 25 (B) 30 (C) 36 (D) 24
- Q.13 $a_2 + a_4 + a_6$ is equal to
(A) 25 (B) 30 (C) 35 (D) 24
- Q.14 Number of such numbers will be
(A) 2 (B) 4 (C) 6 (D) 8

[INTEGER TYPE]

Q.15 to Q.20 are "Integer Type" questions. (The answer to each of the questions are upto 4 digits)

- Q.15 Let $f(x) = x + x^2 + x^4 + x^8 + x^{16} + x^{32} + \dots$ the coefficient of x^{10} in $f(f(x))$ is λ . Find the sum of digits in λ .
- Q.16 If $\left(x + \frac{1}{x} + 1\right)^6 = a_0 + \left(a_1x + \frac{b_1}{x}\right) + \left(a_2x^2 + \frac{b_2}{x^2}\right) + \dots + \left(a_6x^6 + \frac{b_6}{x^6}\right)$, then find the value of a_0 .
- Q.17 Seven different lectures are to be delivered in 7 period of a class on a particular day. Out of 7 lecturers A, B and C are three of them. If the number of ways in which a routine for the day can be made such that 'A' deliver his lecture before 'B' and 'B' before 'C' is N, then find the value of $\left(\frac{N}{120}\right)$.
- Q.18 Let N denotes the number of odd integers between 550 and 800 using the digits 4, 5, 6, 7, 8 and 9. Find the sum of the digits in N.
- Q.19 If $\frac{{}^{11}C_0 {}^{10}C_1}{1} + \frac{{}^{11}C_1 {}^{10}C_2}{2} + \frac{{}^{11}C_2 {}^{10}C_3}{3} + \dots + \frac{{}^{11}C_9 {}^{10}C_{10}}{10} = \left(\frac{{}^{22}C_{10} - a}{b}\right)$ then find the value of $(a + b)$.
- Q.20 If a biased coin is tossed n times such that getting a head is thrice likely to appear than tail. Let P_n be the probability that no two consecutive heads appears. If the recurrence relation is $aP_n = bP_{n-1} + cP_{n-2}$, where $a, b, c \in \mathbb{N}$ then find the least value of $(a + b + c)$.

ANSWER KEY

Q.1	B	Q.2	A	Q.3	C	Q.4	B	Q.5	D
Q.6	C	Q.7	B	Q.8	D	Q.9	C	Q.10	C
Q.11	C	Q.12	C	Q.13	A	Q.14	C	Q.15	4
Q.16	0141	Q.17	7	Q.18	0006	Q.19	0013	Q.20	0023

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